

Notes: Juhász (AER, 2018)

Temporary Protection and Technology Adoption: Evidence from the Napoleonic Blockade

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1 Big picture

- **Question.** Can temporary trade protection help a domestic industry adopt frontier technology and develop enough capacity to persist after protection ends?
- **Setting.** The paper studies mechanized cotton spinning in the French Empire during the Napoleonic Wars and the Continental Blockade, 1803–1815.
- **Natural experiment.** The blockade did not simply shut trade everywhere. It redirected British trade through different smuggling routes, raising the effective distance to London more for some French departments than for others.
- **Treatment.** Treatment intensity is the increase in effective distance to London. This is interpreted as a region-specific increase in protection from British competition.
- **Short-run outcome.** Mechanized cotton spinning capacity, measured by spindles per thousand inhabitants, before the disruption in 1803 and near the end of the blockade in 1812.
- **Short-run finding.** Departments with larger trade-cost shocks increased mechanized spinning capacity substantially more than departments with smaller shocks.
- **Long-run question.** Did temporary protection only create a short-lived protected industry, or did it shift the persistent location of mechanized cotton spinning and industrial development?
- **Long-run design.** The trade-cost shock is used as an instrument for 1812 mechanized spinning capacity when studying later outcomes in 1840, 1887, 1860, 1896, 1930, and 2000.
- **Long-run finding.** Protection-induced 1812 spinning capacity predicts higher mechanized spinning in 1840 and 1887. It also predicts higher industrial value added per capita in 1860, but the aggregate industrial effect fades later.
- **Core message.** Temporary protection can help induce technology adoption and persistent industrial location when underlying technological and market preconditions are present.

2 Incremental contribution of the paper

- **First well-identified reduced-form evidence on infant industry protection.** Earlier infant-industry work often used calibrated counterfactuals or policy episodes where governments chose protected sectors endogenously. This paper exploits within-country regional variation generated by wartime trade disruption, not a deliberate industry-by-region policy choice.
- **A natural experiment in trade protection.** The Napoleonic Blockade created differential increases in the cost of accessing British goods across regions of the French Empire. This delivers continuous treatment intensity similar to a tariff shock but with plausibly exogenous cross-regional variation.
- **Links trade policy to technology adoption, not only trade flows.** The paper moves beyond asking whether protection raises prices or output. It asks whether protection induces adoption of a frontier production technology.
- **Combines short-run adoption with long-run persistence.** It shows both immediate effects on mechanized cotton spinning and persistence in the spatial location of the industry decades later.
- **Builds a new historical dataset.** The paper digitizes industrial surveys of mechanized cotton spindles, reconstructs routes from Lloyd’s List shipping records, and links the blockade to department-level technology adoption.
- **Separates infant-industry mechanisms from general wartime shocks.** Placebos on other sectors, pre-trends, market access, machine quality, and capital-labor ratios are used to test whether the trade-cost shock worked specifically through cotton spinning.

- **Clarifies policy relevance.** The findings support the empirical relevance of infant-industry mechanisms, but the paper is careful that protection may work only when preconditions such as access to knowledge, inputs, and latent capability already exist.

3 Data and measurement

3.1 Historical setting and treatment variation

- Britain was the industrial leader in mechanized cotton spinning. French producers were initially less competitive and relied heavily on British know-how and imported cotton inputs.
- The Continental Blockade attempted to stop British goods from entering Continental Europe.
- Direct trade with France became difficult, but British goods continued to enter through third-country ports and overland routes.
- The key empirical fact is geographic asymmetry: northern France experienced larger increases in effective distance to London than southern France because trade was diverted toward southern routes and overland smuggling.
- The treatment variable is the natural log of effective distance from each department to London before and during the blockade. The shock is the change in this effective distance.

Interpretation: Effective distance works like a region-specific tariff-equivalent shock. A department whose route to British imports became much more circuitous received stronger protection from British competition.

3.2 Short-run outcome: mechanized spinning capacity

- The main outcome is the number of mechanized cotton spindles per thousand inhabitants.
- The paper observes departments in 1803, before the main blockade disruption, and in 1812, near the end of the blockade.
- Spindles are a physical-capital measure and are the standard capacity measure in mechanized cotton spinning.
- Per capita variables use 1811 population to avoid confounding treatment effects with endogenous population changes.
- Some firm-level spindle counts are imputed using predictive mean matching when reports include machines but not spindle counts.

Interpretation: This outcome directly measures adoption of the frontier mechanized spinning technology rather than only output or employment.

3.3 Long-run outcomes

- Mechanized spinning capacity is observed again in 1840 and 1887.
- Industrial value added per capita is observed for 1860, 1896, 1930, and 2000, using historical regional data.
- Export outcomes for cotton manufactures are digitized from primary sources for 1787–1828.
- Additional department characteristics include literacy, market potential, access to knowledge institutions, streams, coal proximity, and lagged spinning capacity.

Interpretation: These outcomes test whether temporary protection created only a wartime sector or generated persistent local industrial development.

4 Models and empirical specifications

4.1 Short-run continuous difference-in-differences

The baseline short-run specification is

$$S_{it} = \alpha_i + \delta_t + \gamma \ln D_{it} + \varepsilon_{it}, \quad (1)$$

where S_{it} is mechanized spinning capacity in department i at time t , $\ln D_{it}$ is effective distance to Britain, α_i are department fixed effects, and δ_t are time fixed effects.

Interpretation: The coefficient γ compares departments with larger versus smaller increases in effective distance before and after the blockade. A positive γ means that stronger protection from British competition caused more domestic mechanized spinning capacity.

4.2 Long-run IV specification

The long-run specification uses 1812 spinning capacity as the endogenous regressor:

$$Y_{it} = \alpha_0 + \beta_{0t} S_i(1812) + \eta_{it}, \quad (2)$$

with first stage

$$S_i(1812) = \alpha_1 + \beta_1 \Delta \ln D_i + \omega_i. \quad (3)$$

Here Y_{it} is a later department-level outcome such as spinning capacity in 1840 or 1887 or industrial value added per capita in later years.

Interpretation: The instrument isolates the part of post-blockade spinning capacity generated by temporary protection. The IV estimand is a local average treatment effect of protection-induced mechanized spinning capacity.

4.3 Reduced form and exclusion logic

Even if the exclusion restriction were imperfect, the reduced form still identifies the effect of the trade-cost shock on later outcomes under the weaker assumption that the shock is uncorrelated with other determinants of those outcomes.

Interpretation: This is important because the long-run horizon is very long. The paper therefore presents long-run IV evidence as suggestive and supports it with placebos and robustness checks.

5 Identification and empirical design

5.1 What identifies the short-run effect

- Treatment intensity comes from changes in possible trade routes, not from fixed geographic distance.
- Department fixed effects absorb permanent regional differences.

- Time fixed effects absorb aggregate changes between 1803 and 1812.
- The comparison is between departments whose effective access to Britain changed more versus less.
- Identification requires no other shock contemporaneous with and correlated with the trade-cost shock.

Interpretation: The design is a continuous-treatment difference-in-differences strategy. It is stronger than a simple north-south comparison because it uses department-level variation in the magnitude of the effective-distance shock.

5.2 Why the setting helps with infant-industry identification

- The protection shock was generated by war and smuggling-route geography, not by a policy maker selecting cotton departments for protection.
- The industry was at the technological frontier and was central to nineteenth-century industrialization.
- The paper has direct measures of technology adoption, not only tariffs or production.
- The short-run period is close to the shock, limiting the number of alternative historical forces that could explain the adoption response.

Interpretation: The paper uses an accidental quasi-experiment to study a question usually confounded by policy endogeneity: whether protection can cause adoption of a frontier industry.

5.3 Main threats and responses

- **Changing market access.** The blockade might have changed access to French or continental markets. Table 2 controls for market potential and alternative market access measures.
- **Pre-existing trends.** Departments that later received larger trade shocks might already have been growing faster in cotton. Table 3 tests pre-treatment cotton growth.
- **Other wartime sectoral shocks.** The shock might proxy for factor-price or regional war shocks. Table 3 uses placebo outcomes in other industries and within cotton spinning.
- **Locational fundamentals.** Long-run persistence may reflect pre-existing advantages. Tables 4–6 include lagged spinning and locational controls.
- **Long-run confounding.** The paper acknowledges that long-run estimates are more suggestive because later outcomes could be affected by other correlated shocks.

Interpretation: The short-run causal case is strongest. The long-run results are informative because they use the same exogenous shock, but they require stronger assumptions over a longer horizon.

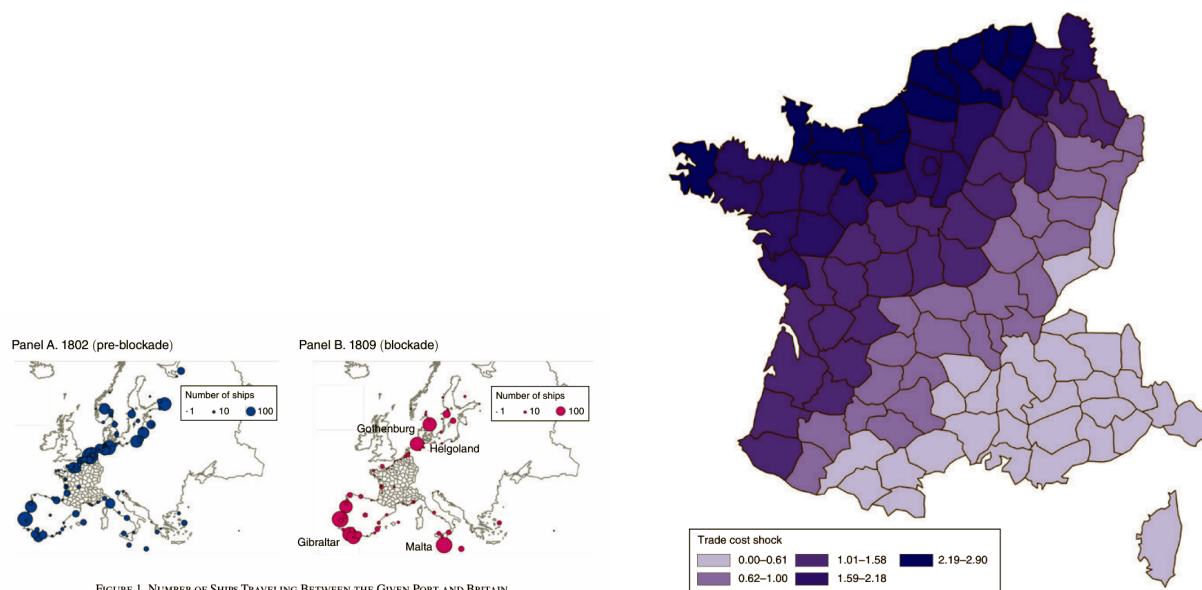
6 Figures and tables

6.1 Figure 1: Number of Ships Traveling Between the Given Port and Britain + Figure 2: Trade Cost Shock

Read:

- Figure 1 compares shipping links with Britain in 1802 and 1809.
- In 1802, shipping between Britain and Continental ports is broad and concentrated in northern European channels.
- By 1809, shipping shifts sharply toward smuggling hubs such as Gothenburg, Helgoland, Gibraltar, and Malta.
- Figure 2 maps the department-level treatment intensity: the log change in effective distance to London.
- Northern and northwestern departments experience the largest trade-cost shocks; southern departments are less affected because access through southern routes remains possible.

Economic message: The two figures together show the first stage of the historical experiment: the blockade redirected trade routes and created spatially unequal protection from British competition.



Source: Lloyd's List

FIGURE 2. TRADE COST SHOCK (*log Change in Effective Distance to London*)

(a) Figure 1: Number of Ships Traveling Between the Given Port and Britain

(b) Figure 2: Trade Cost Shock

6.2 Figure 3: Variation Used, Short-Run Regressions + Table 1: Short-Run Effect of Trade Protection on Mechanized Cotton Spinning Capacity

Read:

- Figure 3 maps spinning capacity in 1803 and 1812. Capacity expands sharply during the blockade, especially along the English Channel and northern departments.

- The largest post-blockade concentration appears in Seine-Maritime, near the region receiving strong protection.
- Table 1 estimates the short-run effect. The coefficient on effective distance is large, positive, and statistically significant across specifications.
- Column 1 gives a coefficient of about 33.47; the standardized coefficient is 0.47.
- Controls for streams, coal, market potential, knowledge access, and literacy do not eliminate the effect.
- The point estimate implies that moving from the 25th to the 75th percentile of the trade-cost shock raises predicted spinning capacity by about the size of mean 1812 spinning capacity.

Economic message: The short-run evidence is the core causal result: stronger temporary protection induces more rapid adoption of mechanized cotton spinning.

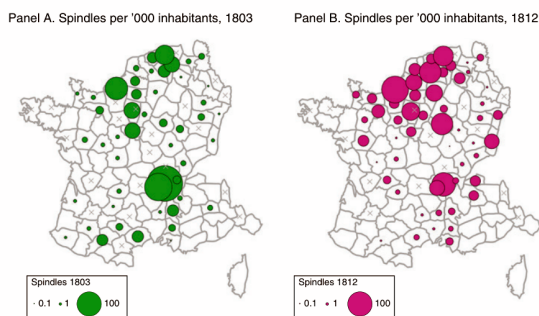


FIGURE 3. VARIATION USED: SHORT-RUN REGRESSIONS

Notes: Missing or dropped observations denoted by X. Departments observed for only one time period not shown to ensure comparability of the graphs across the two time periods. Scale not comparable across time periods.

(a) Figure 3: Variation Used: Short-Run Regressions

	Spindles per thousand inhabitants						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Effective distance	33.47 <i>0.47</i> (9.80)	33.48 <i>0.47</i> (9.89)	34.78 <i>0.49</i> (10.06)	24.73 <i>0.35</i> (10.47)	32.96 <i>0.46</i> (10.90)	42.18 <i>0.52</i> (12.54)	38.82 <i>0.48</i> (13.23)
Streams × 1812		−0.14 (1.50)					−1.16 (2.17)
Coal × 1812			−3.93 (4.21)				4.11 (7.47)
Market potential × 1812				41.05 (21.58)			30.19 (30.19)
Knowledge access × 1812					40.87 (15.22)		34.90 (21.79)
Literacy × 1812						46.41 (21.16)	27.79 (18.86)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	176	176	176	176	176	126	126
Adjusted R^2	0.34	0.33	0.34	0.36	0.37	0.42	0.45
Num. clusters (dept)	88	88	88	88	88	63	63
Num. clusters (gen)	40	40	40	40	40	30	30

Notes: Dependent variable: spindles per thousand inhabitants in department i at time t . Departmental population held constant at its 1811 level. Effective distance is measured as the natural logarithm of the shortest route to London for each department i at time t . Controls (all interacted with an indicator variable which takes the value of one in 1812 and is zero otherwise): Literacy is measured as the proportion of men able to sign their wedding certificate in 1786. Coal is the inverse of log distance to the nearest coalfield. Streams is defined as the natural logarithm of mean streamflow (m³/s). Knowledge access is defined as market access to universities in 1802. Market potential is defined as market access to urban population in 1800. Standardized coefficients in italics. Standard errors clustered at the level of the department in parentheses, standard errors clustered by généralités in curly brackets. The number of observations differ across columns because of missing observations for the literacy measure. For further details on the data, see online Appendix A.3.

(b) Table 1: Short-Run Effect of Trade Protection

6.3 Table 2: Robustness to Changing Market Access + Table 3: Falsification Tests

Read:

- Table 2 asks whether the baseline effect is really driven by changing market access rather than protection from British competition.
- The effective-distance coefficient remains positive and significant after controlling for market potential, time-varying market potential, external market access, and variants excluding Spanish cities.
- Table 3 conducts placebo tests.
- Columns 1–3 show no strong pre-treatment effect on cotton spinning during 1794–1803.
- Columns 4–5 test within cotton spinning and find no evidence that the shock operates through capital-labor ratios or machine quality.
- Columns 6–7 test other industries, wool and leather, and do not show the same pattern.

Economic message: These tables support the exclusion interpretation: the blockade shock appears to affect mechanized cotton spinning adoption rather than simply proxying for general regional trends, market access, or broad wartime industrial changes.

TABLE 2—ROBUSTNESS TO CHANGING MARKET ACCESS

	Spindles per thousand inhabitants					
	(1)	(2)	(3)	(4)	(5)	(6)
Effective distance	33.47 <i>0.47</i> (9.80)	24.73 <i>0.35</i> (10.90)	33.58 <i>0.47</i> (10.15)	40.56 <i>0.57</i> (12.37)	38.50 <i>0.54</i> (10.41)	44.04 <i>0.62</i> (11.36)
Market potential $\times$ 1812		41.05 (21.58)				32.04 (22.55)
Market potential (time var.)			-20.68 (92.70)		-248.90 (136.52)	
Market potential (ext.) $\times$ 1812				40.04 (33.48)		
Market potential (ext. exc. ESP) $\times$ 1812					32.41 (13.38)	59.60 (19.04)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	176	176	176	176	176	176
Adjusted R^2	0.34	0.36	0.33	0.34	0.36	0.37
Num. clusters (dept)	88	88	88	88	88	88
Num. clusters (gen)	40	40	40	40	40	40

Notes: Dependent variable: spindles per thousand inhabitants in department i at time t . Departmental population held constant at its 1811 level. Effective distance is measured as the natural logarithm of the shortest route to London for each department i at time t . Controls: Market potential as defined previously (market access to urban population in 1800). Market potential (time var.) is a time varying measure which takes account of changes in the border of the French Empire between 1803–1812. Market potential (ext.) is market access to urban population in territories outside the boundaries of the French Empire under French influence. Market potential (ext. exc. ESP) drops Spanish cities from the measure because of the insurgency against French rule. All controls except Market potential (time var.) are interacted with a dummy variable that takes the value of one in 1812 and is zero otherwise. Standardized coefficients in italics. Standard errors clustered at the level of the department in parentheses, standard errors clustered by généralités in curly brackets. For further details on the data, see online Appendix A.3.

(a) Table 2: Robustness to Changing Market Access

TABLE 3—FALSIFICATION TESTS

	Pre-treatment period: 1794–1803			Treatment period: 1803–1812			
	Spind. (1)	Spind. (2)	Spind. (3)	K/L (4)	Mach. (5)	Wool (6)	Leather (7)
Effective distance	5.89 <i>0.18</i> (2.94)	3.32 <i>0.10</i> (3.56)	2.08 <i>0.06</i> (4.90)	-0.07 <i>-0.07</i> (0.26)	-0.02 <i>-0.06</i> (0.10)	-2.25 <i>-0.07</i> (2.93)	-0.02 <i>-0.13</i> (0.01)
Market potential $\times$ 1812		12.08 (5.85)	9.47 (8.93)				
Streams $\times$ 1812			-0.10 (0.53)				
Coal $\times$ 1812			2.53 (3.23)				
Knowledge access $\times$ 1812			4.93 (5.74)				
Literacy $\times$ 1812			0.44 (3.33)				
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	176	176	126	78	74	138	116
Adjusted R^2	0.19	0.21	0.15	0.32	0.11	0.18	0.05
Num. clusters (dept)	88	88	63	39	37	69	58
Num. clusters (gen)	40	40	30	23	21	32	28

Notes: Columns 1–3: pre-treatment trends test for mechanized cotton spinning. Columns 4–5: falsification test for other outcome variables in mechanized cotton spinning. Columns 6–7: placebo test for two other industries: wool spinning and leather tanning. Dependent variable in Columns 1–3: number of spindles per thousand inhabitants in department i in 1794 and 1803. Departmental population held constant at its 1811 level. Column 4: capital-labor ratio in mechanized cotton spinning in department i at time t measured as the log number of spindles per unit of labor employed. Column 5: capacity in different vintages of machines measured as the proportion of spindles used in mule jennys relative to spindles in “filatures continues” in department i at time t . Column 6: labor employed in woolen spinning per thousand inhabitants in department i at time t . Employment measured in 1792 and 1811. Column 7: number of leather tanning firms in department i at time t . Number of firms measured in 1794 and 1811. Effective distance is calculated as the natural logarithm of the shortest route to London for each department i at time t . Controls (all interacted with an indicator variable which takes the value of one in 1812 and is zero otherwise): Literacy measured as the proportion of men able to sign their wedding certificate in 1786. Coal is the inverse of log distance to the closest coalfield. Streams is defined as the natural logarithm of mean streamflow (m³/s). Knowledge access is defined as market access to universities in 1802; Market potential is defined as market access to urban population in 1800. The number of observations differ across columns 1–3 because of missing observations for the literacy measure. Columns 4–5 are estimated on the subsample of departments with positive spinning capacity in both 1803 and 1812. The dependent variable is only defined for these departments. Sample size differs across columns as not all departments reported labor employed and the type of machine used. Columns 6–7 are estimated on the largest sample for which the data are available. For further details on the data, see online Appendix A.3. Standardized

(b) Table 3: Falsification Tests

6.4 Figure 4: Variation Used, Long-Run Persistence Regressions + Table 4: Persistence in Cotton Spinning Activity, 1840–1887

Read:

- Figure 4 maps mechanized spinning in 1840 and 1887. Northern departments continue to dominate long after the blockade ends.
- Southern departments stagnate or decline, which is inconsistent with a simple slow diffusion story from Britain to the south.
- Table 4 shows persistence in the location of cotton spinning.
- OLS estimates are positive: one more spindle per thousand inhabitants in 1812 predicts about 2.5–5.1 more spindles in later years depending on controls and year.
- 2SLS estimates using the trade-cost shock as an instrument are also positive and significant, around 2.1–2.7 in 1840 and 4.7–4.9 in 1887.
- Lagged 1803 spinning and locational controls do not remove the persistent effect.

Economic message: Temporary protection changes not only the level of adoption during the shock, but also the long-run geography of the industry.

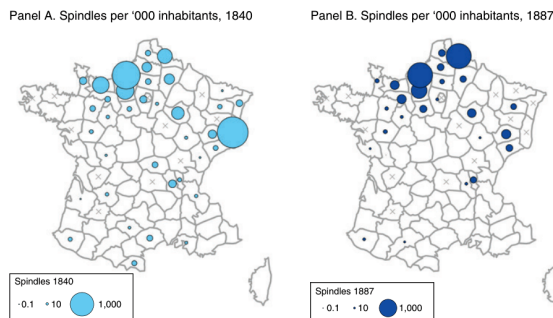


FIGURE 4. VARIATION USED: LONG-RUN PERSISTENCE REGRESSIONS

Notes: Missing or dropped observations are denoted by X. Departments not observed in 1812 not shown as these are missing from the regressions as the regressor of interest is not observed. Haut-Rhin and Bas-Rhin were ceded to Germany 1871–1918. Data for 1887 is not available for these latter departments. Scale not comparable across time periods.

is territory and missing observations for the control variables. Relative to the shock

(a) Figure 4: Variation Used: Long-Run Persistence Regressions

TABLE 4—PERSISTENCE IN THE LOCATION OF COTTON SPINNING ACTIVITY, 1840–1887: OLS AND 2SLS

Dep var measured in:	Spindles per thousand inhabitants							
	OLS				2SLS			
	1840 (1)	1840 (2)	1887 (3)	1887 (4)	1840 (5)	1840 (6)	1887 (7)	1887 (8)
Spindles, 1812	3.04 (0.99) [0.99]	2.47 (0.93) [0.93]	4.75 (1.54) [1.57]	5.06 (1.71) [1.27]	2.12 (1.27) [0.94]	2.68 (0.93) [0.94]	4.72 (1.26) [1.29]	4.85 (1.39)
Spindles, 1803	−2.95 (1.53)	−1.55 (1.01)	−4.69 (2.17)	−4.86 (2.42)	−1.61 (1.56)	−1.85 (1.04)	−4.64 (1.68)	−4.57 (1.84)
Literacy		71.18 (55.35)		−16.63 (92.54)		60.98 (55.34)		−4.29 (95.46)
Market potential		2.67 (109.61)		−33.57 (132.55)		−15.75 (104.61)		−16.38 (129.33)
Knowledge access		−141.21 (83.05)		−159.55 (108.88)		−140.59 (78.31)		−159.00 (101.80)
Coal		−27.19 (20.57)		12.23 (44.00)		−27.43 (18.84)		12.76 (41.19)
Streams		−11.19 (5.80)		−16.85 (10.08)		−10.45 (4.86)		−17.54 (8.34)
Observations	70	63	67	61	70	63	67	61
Adjusted R^2	0.39	0.61	0.61	0.61	12.78	10.35	15.21	10.15
KP F -stat								
Num. clusters (gen)	34	30	33	29	34	30	33	29

Notes: Dependent variable: spindles per thousand inhabitants for the respective year denoted at the top of each column. Departmental population held fixed at its 1811 level across all variables measured in per capita terms. Regressor of interest: spindles per thousand inhabitants in 1812. The instrument is the trade cost shock. Controls: spindles per thousand inhabitants in 1803. Literacy is measured as the proportion of men able to sign their wedding certificate in 1786. Coal is the inverse of log distance to the closest coalfield. Streams is defined as the natural logarithm of mean streamflow (m³/s). Knowledge access is defined as market access to universities in 1802. Market potential is defined as distance to urban population in 1800. All variables measured at their pre-blockade values. The number of observations differ across columns as controls are missing for some departments, while territorial losses to Germany in 1871 account for the difference in observations across the years 1840 and 1887. For further details on the data, see online Appendix A.3. Robust standard errors in parentheses, standard errors clustered by généralités in curly brackets. The latter is not reported in cases where the number of généralités is less than 30.

(b) Table 4: Persistence in Cotton Spinning Activity

6.5 Table 5: Placebo for Persistence in Mechanized Cotton Spinning + Table 6: Industrial Value Added per Capita Outcomes, 1860–2000

Read:

- Table 5 is a placebo for the long-run persistence strategy.
- It asks whether 1812 spinning capacity, instrumented by the trade-cost shock, predicts pre-blockade 1803 spinning capacity.
- The 2SLS coefficients are small and statistically insignificant, supporting the claim that the instrument is not simply picking up pre-existing cotton-spinning locations.
- Table 6 studies aggregate industrial value added per capita.
- OLS estimates are positive in all years; the 2SLS effect is large and significant in 1860 but much smaller and insignificant in later years.
- The industrial value-added effect therefore appears strongest in the medium run, not permanently through 2000.

Economic message: Temporary protection generated persistent sectoral location effects and a medium-run aggregate industrial gain, but not a permanent regional GDP advantage. The strongest evidence is for technology adoption and industry location rather than indefinite broad regional development.

TABLE 5—PLACEBO FOR PERSISTENCE IN THE LOCATION OF MECHANIZED COTTON SPINNING

Dep var measured in:	Spindles per thousand inhabitants							
	OLS		2SLS		First stage		Reduced form	
	1803 (1)	1803 (2)	1803 (3)	1803 (4)	1812 (5)	1812 (6)	1803 (7)	1803 (8)
Spindles 1812	0.28 (0.11) {0.13}	0.31 (0.12) {0.14}	0.03 (0.11) {0.13}	−0.08 (0.27) {0.32}				
Literacy		−17.19 (8.51)		−1.25 (10.07)		24.22 (21.71)		−3.27 (8.04)
Market potential		−14.62 (12.96)		29.29 (34.77)		54.79 (42.93)		24.72 (20.50)
Knowledge access		5.34 (7.77)		8.31 (10.64)		40.16 (27.90)		4.96 (13.56)
Coal		4.38 (5.62)		9.14 (11.91)		12.32 (14.65)		8.11 (9.37)
Streams		1.36 (1.66)		0.30 (1.43)		−0.81 (2.66)		0.37 (1.46)
Trade cost shock					42.68 (16.21) {17.44}	35.82 (20.00) {21.49}	1.31 (5.23) {6.22}	−2.99 (8.83) {10.91}
Observations	71	63	71	63	71	63	71	63
Adjusted R^2	0.40	0.41			0.15	0.22	−0.01	−0.01
KP F -stat			6.937	3.208				
Num. clusters (gen)	34	30	34	30	34	30	34	30

Notes: Dependent variable: spindles per thousand inhabitants by department for the respective year denoted at the top of each column. Departmental population held fixed at its 1811 level across all variables measured in per capita terms. Regressor of interest: spindles per thousand inhabitants in 1812. The instrument is the trade cost shock. Controls: Literacy is measured as the proportion of men able to sign their wedding certificate in 1786. Coal is the inverse of log distance to the closest coalfield. Streams is defined as the natural logarithm of mean streamflow (m³/s). Knowledge access is defined as market access to universities in 1802. Market potential defined as distance to urban population in 1800. The number of observations differ across columns as controls are missing for some departments. Robust standard errors in parentheses, standard errors clustered by généralités in curly brackets.

(a) Table 5: Placebo for Persistence

TABLE 6—INDUSTRIAL VALUE ADDED PER CAPITA OUTCOMES, 1860-2000: OLS AND 2SLS

Dep var measured in:	Natural logarithm of industrial value added per capita							
	OLS				2SLS			
	1860 (1)	1896 (2)	1930 (3)	2000 (4)	1860 (5)	1896 (6)	1930 (7)	2000 (8)
Spindles, 1812	0.0037 <i>0.3925</i> (0.0012) {0.0013}	0.0025 <i>0.2394</i> (0.0012) {0.0013}	0.0040 <i>0.3965</i> (0.0016) {0.0016}	0.0025 <i>0.2527</i> (0.0012) {0.0012}	0.0075 <i>0.7987</i> (0.0026) {0.0021}	0.0010 <i>0.0937</i> (0.0026) {0.0025}	0.0016 <i>0.1590</i> (0.0030) {0.0028}	0.0031 <i>0.3128</i> (0.0026) {0.0025}
Spindles, 1803	0.0035 (0.0020)	0.0048 (0.0019)	0.0046 (0.0020)	0.0053 (0.0017)	−0.0020 (0.0035)	0.0070 (0.0036)	0.0081 (0.0041)	0.0044 (0.0032)
Observations	68	66	68	68	68	66	68	68
Adjusted R^2	0.2414	0.1369	0.2772	0.1718				
KP F -stat					12.60	15.25	12.60	12.60
Num. clusters (gen)	33	32	33	33	33	32	33	33

Notes: Dependent variable: Natural logarithm of industrial value added per capita measured at the level of the department. Departmental population held fixed at its 1811 level across all variables measured in per capita terms. Regressor of interest: spindles per thousand inhabitants in 1812. The instrument is the trade cost shock. Standardized coefficient in italics. The number of observations differ across columns because of territorial losses to Germany between 1871–1919. For further details on the data, see online Appendix A.3. Robust standard errors in parentheses, standard errors clustered by généralités in curly brackets.

(b) Table 6: Industrial Value Added per Capita Outcomes

7 Short summary

- The paper studies whether temporary protection can induce technology adoption in an infant industry.
- The setting is mechanized cotton spinning in the French Empire during the Napoleonic Wars.
- The Continental Blockade redirected trade rather than stopping it completely, generating regionally different increases in effective distance to Britain.
- This creates a continuous treatment measure of protection from British competition.
- The short-run design compares departments with larger versus smaller trade-cost shocks between 1803 and 1812.
- Treated departments increase mechanized cotton spinning capacity substantially more.
- The results are robust to controls for market access, streams, coal, knowledge access, and literacy.
- Placebo tests show no comparable pre-trend or similar effects in other industries.
- The long-run analysis instruments 1812 spinning capacity with the trade-cost shock.
- Protection-induced spinning capacity persists in the location of cotton spinning in 1840 and 1887.
- Industrial value added per capita rises in 1860, but the effect fades in later years.
- The paper is careful that long-run evidence is more suggestive because many other historical forces can operate over decades.
- The evidence supports the empirical relevance of infant-industry mechanisms when underlying capabilities are already present.

8 Conclusion

8.1 Core conclusion

Juhász provides evidence that temporary protection can cause adoption of frontier technology in an infant industry. The Napoleonic Blockade accidentally protected some French departments more than others by

raising the effective cost of importing British yarn. Regions that became more protected adopted mechanized cotton spinning more intensively, and the post-blockade industry persisted in those locations for decades.

8.2 Mechanism and interpretation

The key economic mechanism is that British competition initially made mechanized cotton spinning unprofitable for many French entrepreneurs. Once the blockade raised the price of competing British yarn, adoption became profitable in departments receiving large trade-cost shocks. The persistence results are consistent with agglomeration or learning-by-doing: early adopters gained experience, supplier linkages, worker skills, or local knowledge that made production profitable even after the temporary shock ended.

8.3 Incremental contribution for the infant-industry debate

The paper's incremental contribution is to make the infant-industry argument empirically testable in a setting where policy endogeneity is minimized. Rather than observing a government choosing a sector and then asking whether the policy worked, the paper observes a war-induced shock that changed protection unevenly across regions within the same empire. This makes it possible to compare otherwise similar regions that faced different exposure to British competition.

8.4 Policy implications

The results do not imply that protection is generally welfare improving. The paper emphasizes that many prerequisites were already present in France: entrepreneurs, knowledge access, machinery know-how, industrial surveys, and a latent capability to produce. The results suggest that temporary protection can help when market failures such as learning-by-doing or agglomeration externalities prevent a potentially competitive industry from emerging. But when these preconditions are absent, protection may be a blunt and costly tool.

8.5 Limitations

The short-run evidence is relatively strong because it is close to the historical shock and supported by placebos. The long-run evidence is more suggestive because subsequent decades include many other shocks, policies, and market changes. The paper also cannot perfectly separate different agglomeration mechanisms, such as learning-by-doing, supplier linkages, labor pooling, or knowledge spillovers. Finally, the regional evidence shows persistent industry location and temporary aggregate industrial gains, but it cannot fully resolve the welfare cost of protection or compare it to alternative policies.

8.6 Takeaway

The paper turns a classic theoretical claim into a credible historical empirical test. Temporary protection can shape the adoption and geography of frontier industries, but its success depends on underlying capabilities and the existence of dynamic externalities. The most important lesson is not that protection always works; it is that the infant-industry mechanism can be real under the right conditions.